

 OR-AS Operations Research Applications and Solutions	Case Name: Railway Bridge (1)	Sector	Construction (civil)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Mieke De Saedeleer	Project Control	User defined distributions
Date	2016		Automatic tracking
File Name	C2016-01 Railway Bridge (1)		Tracking based on user input

1. Project description

Project authenticity

The project involves the demolition of an existing railway bridge and the construction of a new multi-phase reinforced concrete railway bridge, including preparatory studies, approvals, foundation works, structural phases, and final road and finishing works.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	31	Serial/Parallel (SP)	51%
Planned Duration (PD)	225	Activity Distribution (AD)	71%
Budget At Completion (BAC)	671,383.50 €	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	86%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	13	15	1.81
CRI-rho	22	19	0.86
CRI-tau	26	36	1.53

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	100	100	0.00
SI	5	15	4.88
SSI	9	17	3.04
CRI-r	13	17	2.54
CRI-rho	17	19	1.83
CRI-tau	16	26	2.62

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	12.33	-0.9	1	0.52	-0.25
PV - SPI	30.95	30.81	CPI	1.23	-1.17
PV - SCI	29.15	28.88	SPI	17	17
ED - 1	18.05	1.33	SPI(t)	17.64	17.64
ED - SPI	30.95	30.81	SCI	16.73	16.73
ED - SCI	30.42	30.27	SCI(t)	17.4	17.4
ES - 1	5.28	-4	0.8 CPI + 0.2 SPI	8.18	8.18
ES - SPI(t)	30.72	30.15	0.8 CPI + 0.2 SPI(t)	8.72	8.71
ES - SCI(t)	30.24	29.67			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 14 tracking. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-25234.29	-65356.95	-34.24	0.93	0.83	0.77	1
std dev	12821.59	57668.67	17.9	0.05	0.16	0.13	0
final	-32320	0	-62	0.95	1	0.78	1

2.3.3.2. Time forecasting

PD	225	Real Duration	274	Late/early	21.78%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	247.36	19.44	9.94	-9.94
PV - SPI	285.15	61.18	26.73	-26.73
PV - SCI	312.77	87.06	39.01	-39.01
ED - 1	253.85	17.55	12.82	-12.82
ED - SPI	290.15	57.54	28.96	-28.96
ED - SCI	306.62	78.37	36.27	-36.27
ES - 1	257.23	16.65	14.32	-14.32
ES - SPI(t)	302.38	54.56	34.39	-34.39
ES - SCI(t)	319.38	74.32	41.95	-41.95

2.3.3.3. Cost forecasting

BAC	671,383.50 €	Real Cost	703,703.50 €	Over/under budget	4.81 %
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	698558.88	10997.57	0.79	0.73
CPI	730806.81	40953.58	4.56	-3.85
SPI	806566.64	147183.87	15.32	-14.62
SPI(t)	828987.9	150623.07	18.51	-17.8
SCI	856228.5	213856.78	22.38	-21.67
SCI(t)	879758.36	212893.38	25.73	-25.02
0.8 CPI + 0.2 SPI	742518.17	56201.19	6.22	-5.52
0.8 CPI + 0.2 SPI(t)	745799.49	55018.22	6.69	-5.98